

Grapes

LAYOUT SPECIMENS

The same passage, or part of it, is set five ways. Each specimen tests a way to carry notes on notes.

1. Stacked footnotes. One column. Numbered notes at the foot; notes on those notes, lettered, below them; notes on those, marked with symbols and run together, at the very bottom.

2. Sidenotes, their notes at the foot. Numbered notes in a wide outer margin. Notes on a sidenote drop to the foot of the page.

3. Sidenotes, their notes in the margin. As 2, but the notes on a sidenote stay under it in the margin, in smaller type.

4. A Talmud-style page. A short main text at the top centre, two commentaries wrapped around it in two typefaces, and a band of notes on the commentaries at the foot. Composed by hand, one page at a time.

5. Notes keyed to line numbers. The layout of a scholarly edition. The text carries no marks; three series of notes at the foot cite the line number and repeat the words they comment on.

The prose is placeholder. Its facts come from memory and have not been checked against sources.

ONE

The Berry

A grape is a berry. Botanists mean something exact by that word: a fleshy fruit that grows from a single flower with a single ovary, its seeds buried in the pulp. By this rule the tomato is a berry, and so are the banana and the eggplant, while the strawberry and the raspberry are not.¹ The grape qualifies without argument.

Nearly all the grapes we eat, dry, and press come from one species, *Vitis vinifera*, the wine-bearing vine.² The rest of the genus, several

¹The strawberry is an accessory fruit. Most of its flesh grows from the receptacle, the part of the flower that holds the ovaries, and the specks on the outside are the true fruits, each one an achene.^a The raspberry is an aggregate: a cluster of many small fruits, each from a separate ovary of one flower.

²*Vinifera* is Latin for wine-bearing, from *vinum*, wine, and *ferre*, to bear. The English word *grape* took a stranger road. It comes from Old French *grape*, a bunch of grapes, and probably before that from a Germanic word for a hook.^b So the name of the fruit first meant the tool, then the cluster, and only last the single berry.

^aAn achene is a small dry fruit with one seed that does not split open when ripe. A sunflower seed in its striped shell is also an achene.*

^bPresumably the hook used to cut the bunches. The same root is thought to lie behind *grapple* and French *grappin*.

*Strictly, botanists call the sunflower's fruit a cypsela, a name reserved for achene-like fruits of the daisy family. Not every author bothers with the distinction.

dozen species, grows wild across the temperate Northern Hemisphere, and a good number of them are American. For most of history this mattered to no one in Europe.

Then, in the 1860s, vines in the south of France began to die.³ Over the next decades the blight moved through France and into the rest of Europe. The cure, found after much argument, was to graft European vines onto American roots. Most of the world's wine grapes still grow that way, a European plant standing on American feet.⁴

The Americans, meanwhile, had grapes of their own. In the 1840s Ephraim Wall Bull, in Concord, Massachusetts, began planting seeds of a wild native species, *Vitis labrusca*, and selecting among the seedlings. The Concord grape came out of that work.⁵

In the Tokaj hills of north-eastern Hungary, the growers wait for a mould. Late in a good autumn, a fungus pierces the skins of the ripe berries and the water inside evaporates. What remains shrivels into something between a raisin and a jewel, and it makes one of the great

³The cause was an insect smaller than a pinhead, now called *Daktulosphaira vitifoliae* but known to everyone by its older name, phylloxera.^a It had come from North America, probably on vines brought over by collectors. American vines had lived with it for a long time and tolerated it. European vines had not.

⁴The exceptions are places the insect never reached or cannot live, such as vineyards on pure sand.^b Chile is the example usually given.

⁵Its flavour, which Europeans tend to call foxy, is the flavour most Americans know as grape: the flavour of grape jelly and grape soda. In 1869 Thomas Welch, a dentist in Vineland, New Jersey, heated Concord juice to stop it from fermenting, and so made an unfermented wine for church communion.^c

^aFrom Greek *phyllon*, leaf, and *xeros*, dry. The name describes the damage the insect does to leaves, although the fatal damage is to the roots.

^bSand seems to hinder the insect's movement through the soil. Why, exactly, is a question I have not yet looked into.*

^cWelch was a supporter of temperance. He was applying a method that Louis Pasteur had worked out only a few years earlier.[†]

*Note to self: look into it.

[†]Pasteur's work on heating wine to prevent spoilage dates from the mid-1860s.

sweet wines.⁶

⁶The fungus is *Botrytis cinerea*, noble rot. The shrivelled berries are called *aszú*.^a
In the wrong weather the same fungus is simply grey mould, the grey fur on a forgotten punnet of strawberries.^b

^aThe word is related to the Hungarian verb for drying up.

^bHere the strawberry returns, as it will throughout this book.*

*This is a promise, not a threat.

ONE

The Berry

A grape is a berry. Botanists mean something exact by that word: a fleshy fruit that grows from a single flower with a single ovary, its seeds buried in the pulp. By this rule the tomato is a berry, and so are the banana and the eggplant, while the strawberry and the raspberry are not.¹ The grape qualifies without argument.

Nearly all the grapes we eat, dry, and press come from one species, *Vitis vinifera*, the wine-bearing vine.² The rest of the genus, several dozen species, grows wild across the temperate Northern Hemisphere, and a good number of them are American. For most of history this mattered to no one in Europe.

Then, in the 1860s, vines in the south of France began to die.³ Over the next decades the blight moved through France and into the rest of Europe. The cure, found after much argument, was to graft European vines onto American roots. Most of the world's wine grapes still grow that way, a European plant standing on American feet.⁴

¹An achene is a small dry fruit with one seed that does not split open when ripe. A sunflower seed in its striped shell is also an achene.*

²Presumably the hook used to cut the bunches. The same root is thought to lie behind *grapple* and French *grappin*.

³From Greek *phyllon*, leaf, and *xeros*, dry. The name describes the damage the insect does to leaves, although the fatal damage is to the roots.

*Strictly, botanists call the sunflower's fruit a cypsela, a name reserved for achene-like fruits of the daisy family. Not every author bothers with the distinction.

¹ The strawberry is an accessory fruit. Most of its flesh grows from the receptacle, the part of the flower that holds the ovaries, and the specks on the outside are the true fruits, each one an achene.^a The raspberry is an aggregate: a cluster of many small fruits, each from a separate ovary of one flower.

² *Vinifera* is Latin for wine-bearing, from *vinum*, wine, and *ferre*, to bear. The English word *grape* took a stranger road. It comes from Old French *grape*, a bunch of grapes, and probably before that from a Germanic word for a hook.^b So the name of the fruit first meant the tool, then the cluster, and only last the single berry.

³ The cause was an insect smaller than a pinhead, now called *Daktulosphaira vitifoliae* but known to everyone by its older name, phylloxera.^c It had come from North America, probably on vines brought over by collectors. American vines had lived with it for a long time and tolerated it. European vines had not.

⁴ The exceptions are places the insect never reached or cannot live, such as vineyards on pure sand.^d Chile is the example usually given.

⁵ Its flavour, which Europeans tend to call foxy, is the flavour most Americans know as grape: the flavour of grape jelly and grape soda. In 1869 Thomas Welch, a dentist in Vineland, New Jersey, heated Concord juice to stop it from fermenting, and so made an unfermented wine for church communion.^a

⁶ The fungus is *Botrytis cinerea*, noble rot. The shrivelled berries are called *aszú*.^b In the wrong weather the same fungus is simply grey mould, the grey fur on a forgotten punnet of strawberries.^c

The Americans, meanwhile, had grapes of their own. In the 1840s Ephraim Wall Bull, in Concord, Massachusetts, began planting seeds of a wild native species, *Vitis labrusca*, and selecting among the seedlings. The Concord grape came out of that work.⁵

In the Tokaj hills of north-eastern Hungary, the growers wait for a mould. Late in a good autumn, a fungus pierces the skins of the ripe berries and the water inside evaporates. What remains shrivels into something between a raisin and a jewel, and it makes one of the great sweet wines.⁶

^dSand seems to hinder the insect's movement through the soil. Why, exactly, is a question I have not yet looked into.*

^aWelch was a supporter of temperance. He was applying a method that Louis Pasteur had worked out only a few years earlier.[†]

^bThe word is related to the Hungarian verb for drying up.

^cHere the strawberry returns, as it will throughout this book.[‡]

*Note to self: look into it. [†]Pasteur's work on heating wine to prevent spoilage dates from the mid-1860s. [‡]This is a promise, not a threat.

ONE

The Berry

A grape is a berry. Botanists mean something exact by that word: a fleshy fruit that grows from a single flower with a single ovary, its seeds buried in the pulp. By this rule the tomato is a berry, and so are the banana and the eggplant, while the strawberry and the raspberry are not.¹ The grape qualifies without argument.

Nearly all the grapes we eat, dry, and press come from one species, *Vitis vinifera*, the wine-bearing vine.² The rest of the genus, several dozen species, grows wild across the temperate Northern Hemisphere, and a good number of them are American. For most of history this mattered to no one in Europe.

Then, in the 1860s, vines in the south of France began to die.³ Over the next decades the blight moved through France and into the rest of Europe. The cure, found after much argument, was to graft European vines onto American roots. Most of the world's wine grapes still grow that way, a European plant standing on American feet.⁴

The Americans, meanwhile, had grapes of their own. In the 1840s Ephraim Wall Bull, in Concord, Massachusetts, began planting seeds of a wild native species, *Vitis labrusca*, and selecting among the seedlings. The Concord grape came out of that work.⁵

In the Tokaj hills of north-eastern Hungary, the growers wait for a mould. Late in a good autumn, a fungus pierces the skins of the ripe berries and the water inside evaporates. What remains shrivels into

¹ The strawberry is an accessory fruit. Most of its flesh grows from the receptacle, the part of the flower that holds the ovaries, and the specks on the outside are the true fruits, each one an achene.^a The raspberry is an aggregate: a cluster of many small fruits, each from a separate ovary of one flower.

^a An achene is a small dry fruit with one seed that does not split open when ripe. A sunflower seed in its striped shell is also an achene.*

* Strictly, botanists call the sunflower's fruit a cypsela, a name reserved for achene-like fruits of the daisy family. Not every author bothers with the distinction.

² *Vinifera* is Latin for wine-bearing, from *vinum*, wine, and *ferre*, to bear. The English word *grape* took a stranger road. It comes from Old French *grape*, a bunch of grapes, and probably before that from a Germanic word for a hook.^a So the name of the fruit first meant the tool, then the cluster, and only last the single berry.

^a Presumably the hook used to cut the bunches. The same root is thought to lie behind *grapple* and French *grappin*.

³ The cause was an insect smaller than a pinhead, now called *Daktulosphaira vitifoliae* but known to everyone by its older name, phylloxera.^a It had come from North America, probably on vines brought over by collectors. American vines had lived with it for a long time and tolerated it. European vines had not.

^a From Greek *phyllon*, leaf, and *xeros*, dry. The name describes the damage the insect does to leaves, although the fatal damage is to the roots.

⁴ The exceptions are places the insect never reached or cannot live, such as vineyards on pure sand.^a Chile is the example usually given.

^a Sand seems to hinder the insect's movement through the soil. Why, exactly, is a question I have not yet looked into.*

* Note to self: look into it.

⁵ Its flavour, which Europeans tend to call foxy, is the flavour most Americans know as grape: the flavour of grape jelly and grape soda. In 1869 Thomas Welch, a dentist in Vineland, New Jersey, heated Concord juice to stop it from fermenting, and so made an unfermented wine for church communion.^a

^a Welch was a supporter of temperance. He was applying a method that Louis Pasteur had worked out only a few years earlier.*

* Pasteur's work on heating wine to prevent spoilage dates from the mid-1860s.

⁶ The fungus is *Botrytis cinerea*, noble rot. The shrivelled berries are called *aszú*.^a In the wrong weather the same fungus is simply grey mould, the grey fur on a forgotten punnet of strawberries.^b

^a The word is related to the Hungarian verb for drying up.

^b Here the strawberry returns, as it will throughout this book.*

* This is a promise, not a threat.

something between a raisin and a jewel, and it makes one of the great sweet wines.⁶

A grape. The single berry, not the bunch. The English word comes from French *grappe*, the bunch, so that the name of the cluster became the name of the berry.

Is a berry. In the botanist's sense, not the grocer's. The grocer calls a berry anything small, soft, and sweet that can be eaten without peeling.

Something exact. The definition here is the textbook one. Botanists argue about the hard cases. **A single flower with a single ovary.** This separates a true

berry from an aggregate fruit such as the raspberry, which grows from one flower with many ovaries, and from a multiple fruit such as the mulberry or the pineapple, which grows from many flowers fused together. **Seeds buried in the pulp.** Compare the stone fruits, the cherry and the plum, whose single seed is locked in a hard pit. **The tomato.** The Supreme Court of the United States once ruled that the tomato is a vegetable, in the case of *Nix v. Hedden*, decided in 1893.^{*} The court did not dispute the botany. It held that in ordinary speech, and therefore in the law, tomatoes are vegetables. **The banana.** The seeds of wild bananas are large and hard. The small dark specks in a cultivated banana are the remains of seeds that no longer develop.[†] **The eggplant.** Like the tomato, a member of the nightshade family. **The strawberry.** Its flesh grows from the receptacle, the part of the flower that holds the ovaries, and the specks on its skin are the true fruits. **Qualifies without argument.** Which is more than can be said of most things in this book. ***Vitis vinifera*.** *Vitis* is the Latin for vine, and *vinifera* means wine-bearing. The rest of the genus, several dozen species, grows wild across the temperate Northern Hemisphere. Many of them are American, and this fact will matter a great deal later.

A grape is a berry. Botanists mean something exact by that word: a fleshy fruit that grows from a single flower with a single ovary, its seeds buried in the pulp. By this rule the tomato is a berry, and so are the banana and the eggplant, while the strawberry and the raspberry are not. The grape qualifies without argument. Nearly all the grapes we eat, dry, and press come from one species, *Vitis vinifera*, the wine-bearing vine.

On this page. The page imitates, loosely, the printed Talmud, where a short passage of main text sits in the centre and the commentaries wrap around it. The main text is set first. Each commentary is then set as one long paragraph whose first lines are narrowed to leave room for it; below the main text the commentaries widen to fill the half page. **Two commentaries.** Here the left column glosses single words, and this column

makes larger remarks. A book about grapes might give the left column to botany and the right to history, or the left to fact and the right to digression. **Keyed by lemma.** The commentaries use no note numbers. Each comment opens with the words it comments on, in bold, and the reader finds the place by matching words. The main text stays clean. The reader pays for this with a little searching. **Two typefaces.** The left column is set in EB Garamond and this one in Linux Libertine, so the eye can tell them apart without labels. Printed Talmuds do something similar: the main text and the commentaries are set in different Hebrew scripts.[‡] **The band at the foot.** It holds a third level: notes on the commentaries, marked with symbols. **What this costs.** Each page must be composed as a unit. The main text for a page has to be chosen by hand, and the commentaries must be long enough to fill the page but not longer. For a whole book this would need either a great deal of hand work or a small program that measures the text and cuts it into pages.

^{*} The tariff at issue, from 1883, taxed imported vegetables but not fruit. [†] Most bananas sold in Europe and North America belong to one group of cultivars, the Cavendish. [‡] The commentaries are traditionally set in what is called Rashi script, a semi-cursive Hebrew typeface. The name is a later convention; Rashi himself did not write in it.

ONE The Berry

A grape is a berry. Botanists mean something exact by that word: a fleshy fruit that grows from a single flower with a single ovary, its seeds buried in the pulp. By this rule the tomato is a berry, and so are the banana and the eggplant, while the strawberry and the raspberry are not. The grape qualifies without argument.

5

Nearly all the grapes we eat, dry, and press come from one species, *Vitis vinifera*, the wine-bearing vine. The rest of the genus, several dozen species, grows wild across the temperate Northern Hemisphere, and a good number of them are American. For most of history this mattered to no one in Europe. The English word *grape* first meant the cluster, and only later the single berry.

10

GLOSS

- 1 **berry**] In the botanist's sense, not the grocer's.
2 **single flower with a single ovary**] This separates a true berry from an aggregate fruit such as the raspberry.
4 **strawberry**] An accessory fruit: its flesh grows from the receptacle.

WORDS

- 7 ***Vitis vinifera***] Latin *vinum*, wine, and *ferre*, to bear.
10 ***grape***] From Old French *grape*, a bunch of grapes; probably from a Germanic word for a hook.

ASIDE

- 3 **tomato**] A United States court once ruled it a vegetable, for the purposes of a tariff.
4 **strawberry**] The strawberry will return throughout this book.
9 **American**] This will matter when phylloxera arrives in France in the 1860s.